NLP Mini Project: Fine-Tuning BERT for Custom Intent Classification using the SNIPS NLU Dataset

A Project Report

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Abstract

This report presents a comprehensive study on fine-tuning a transformer-based model for intent classification using the SNIPS Natural Language Understanding benchmark dataset. The work focuses on adapting a pre-trained BERT model to classify user utterances into predefined intent categories. The performance of the model is evaluated using multiple metrics including accuracy, precision, recall, and F1-scores, along with a detailed analysis using confusion matrices and learning curves.

1 Introduction

1.1 Background

Intent classification is a core component of Natural Language Understanding systems, where the objective is to map a user's textual input to a predefined intent category. This task is crucial in applications such as conversational agents, virtual assistants, and automated customer support systems. Traditional machine learning approaches relied heavily on handcrafted features; however, the emergence of transformer-based architectures has significantly improved performance by leveraging contextual representations.

1.2 Motivation

Pre-trained language models such as BERT have demonstrated remarkable capability in capturing semantic and contextual relationships in text. Fine-tuning such models on domain-specific datasets allows them to adapt to task-specific distributions, thereby improving classification accuracy. The SNIPS dataset, being diverse and well-annotated, serves as an ideal benchmark for evaluating such approaches.

1.3 Objective

The primary objective of this study is to fine-tune a BERT-based model for intent classification and evaluate its performance using standard Natural Language Processing metrics. Additionally, the study aims to analyze training dynamics through loss and metric curves and to identify potential sources of classification errors.

2 Methodology

2.1 Dataset Description and Preprocessing

The dataset used in this study is derived from the SNIPS Natural Language Understanding (NLU) benchmark, which is originally provided in a structured JSON format designed for joint intent detection and slot-filling tasks. In its native form, the dataset is organized hierarchically, where each intent is represented as a key in a dictionary and is associated with a list of annotated utterances.

Each utterance in the dataset is not stored as a single plain-text sentence but rather as a sequence of segments under a `data` field. Each segment contains a piece of text and may optionally include entity annotations. These annotations specify slot-related information such as the entity type and its corresponding value within the utterance.

This segmented representation allows the dataset to encode both the raw text and its semantic structure, making it suitable for tasks involving both intent classification and slot extraction.

An example utterance in the JSON format can be conceptually represented as a sequence of text fragments, where certain fragments are tagged with entity labels. These fragments must be concatenated to reconstruct the complete natural language sentence before being used for downstream tasks. Consequently, the original dataset exhibits a hierarchical structure of the form: intent $\rightarrow$ list of utterances $\rightarrow$ list of text segments.

In this work, the focus is restricted to the intent classification task. Therefore, the dataset is preprocessed to transform the hierarchical JSON structure into a flat representation consisting of sentence–intent pairs. During preprocessing, all text fragments within each utterance are concatenated in sequence to form a complete sentence, while entity annotations are discarded. This transformation simplifies the problem into a single-label multi-class classification task, which is compatible with transformer-based models such as BERT.

After preprocessing, the dataset consists of a collection of 328 natural language utterances paired with their corresponding intent labels. There are 11 unique intent classes. Each instance is independent and contains exactly one intent, making it suitable for supervised learning using standard classification objectives. The dataset contains multiple intent categories corresponding to common user actions, such as querying weather information, playing music, or booking services.

Although the original SNIPS dataset includes rich slot-level annotations, these are not utilized in the current study. However, their presence in the original data contributes to the linguistic diversity and realism of the utterances, which in turn enhances the complexity of the intent classification task. Additionally, certain intent classes exhibit semantic overlap, introducing ambiguity that challenges the model's ability to distinguish between closely related user queries.

Overall, the preprocessing step converts the structured SNIPS dataset into a form suitable for deep learning-based text classification, while retaining the semantic richness necessary for evaluating contextual language understanding models. The preprocessing pipeline involves converting raw text into a format suitable for transformer-based models. This includes tokenization using a BERT tokenizer, padding and truncation to maintain uniform sequence lengths, and encoding intent labels into numerical form. The processed data is then split into training and validation sets to enable model evaluation during training.

2.2 Model Architecture

The model architecture is built upon a pre-trained BERT encoder, which serves as the backbone for extracting contextual representations from input text. The encoder processes tokenized input sequences and produces a high-dimensional embedding corresponding to the special [CLS] token, which captures the overall semantic meaning of the input utterance.

On top of the BERT encoder, a task-specific classification head is added to perform intent prediction. This classification head consists of a fully connected linear layer that maps the [CLS] embedding to a vector of logits corresponding to the number of intent classes. A softmax function is subsequently applied to obtain class probabilities.

To balance computational efficiency and generalization performance, the BERT model is partially fine-tuned. Specifically, the majority of the lower encoder layers are kept frozen, while only the top two encoder layers along with the classification head are updated during training. This design allows the model to retain general linguistic knowledge learned during pre-training while adapting higher-level representations to the target task.

Freezing the lower layers reduces the risk of overfitting, particularly given the moderate size of the dataset, and significantly lowers computational cost. At the same time, fine-tuning the upper layers enables the model to specialize in capturing task-specific semantic distinctions required for accurate intent classification.

2.3 Training Strategy

The model is trained using a supervised learning approach with cross-entropy loss as the objective function. The AdamW optimizer is employed for efficient gradient-based optimization, along with appropriate learning rate scheduling. During training, multiple performance metrics such as accuracy, precision, recall, and F1-score are computed for both training and validation sets to monitor learning progress and detect overfitting.

Additionally, selective fine-tuning is employed as a regularization strategy. By restricting weight updates to only the upper layers of the BERT encoder and the classification head, the model avoids excessive parameter updates that could lead to overfitting. This approach also stabilizes training and allows faster convergence, as fewer parameters are optimized compared to full fine-tuning.

2.4 Evaluation Metrics

To ensure a comprehensive evaluation, multiple metrics are considered. Accuracy provides an overall measure of correct predictions, while precision and recall offer insights into class-wise performance. Both macro and micro averaged F1-scores are computed to account for class imbalance and overall classification quality. Additionally, a confusion matrix is used to visualize prediction errors across different intent classes.

3 Results

3.1 Training Dynamics

The training and validation loss curves, illustrated in Figure 1, reveal important insights into the learning behavior of the model. Initially, both training and validation loss decrease rapidly, indicating that the model effectively captures underlying patterns in the dataset during the early stages of training. However, after 112 epochs, the training loss continues to decrease towards near-zero values, while the validation loss stabilizes and gradually begins to increase. This divergence between the training and validation loss curves is a clear indication of overfitting, where the model starts memorizing the training data rather than generalizing to unseen examples. Hence, the final model was saved on 112th epoch. The later models were not saved due to severe overfitting.

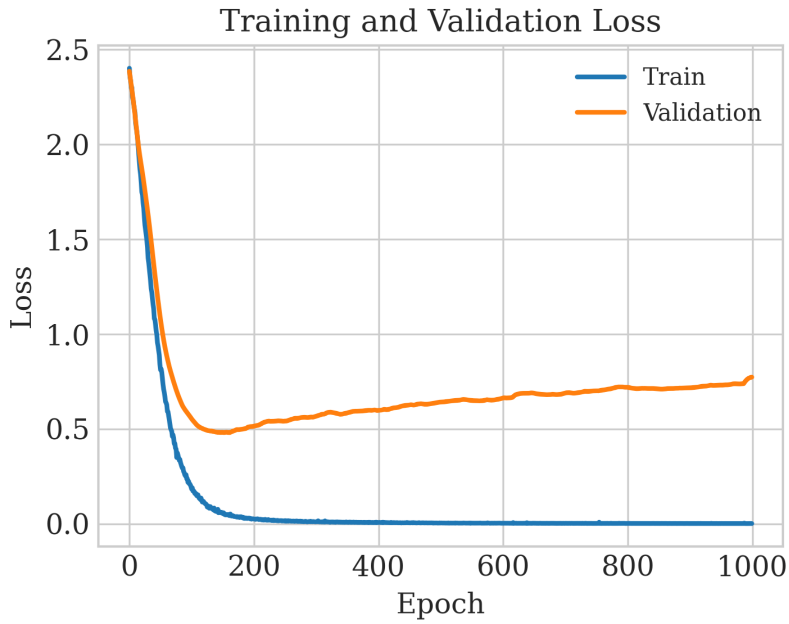

Figure 1: Training and validation loss curves showing convergence and overfitting behavior.

The accuracy trends, shown in Figure 2, further support this observation. The training accuracy increases rapidly and reaches nearly perfect performance, approaching 100% accuracy. In contrast, the validation accuracy improves quickly in the initial phase but plateaus around approximately 83–84%. The persistent gap between training and validation accuracy indicates that while the model has high capacity, it does not fully generalize to unseen data, reinforcing the evidence of overfitting observed in the loss curves.

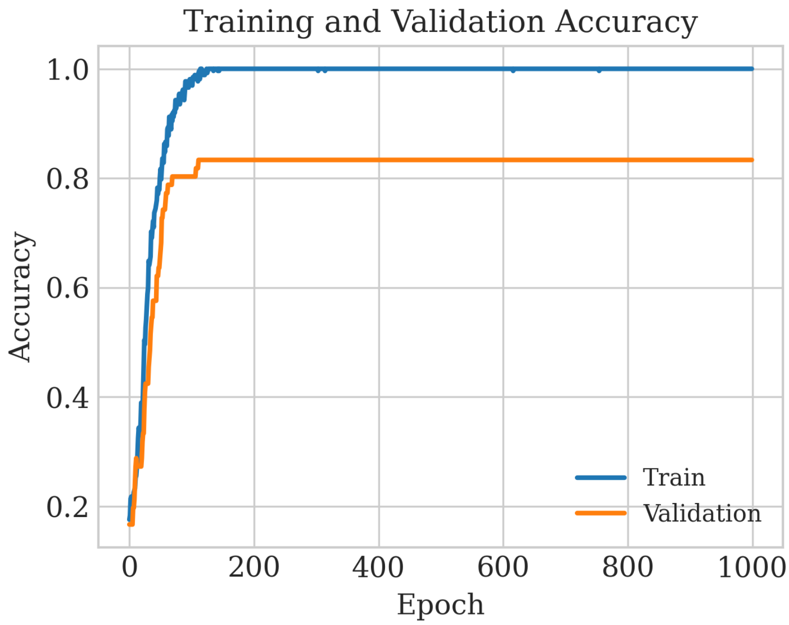

Figure 2: Training and validation accuracy curves demonstrating rapid convergence and generalization gap.

3.2 Evaluation Metric Analysis

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

Accuracy measures the overall proportion of correct predictions. It is easy to interpret but can be misleading in the presence of class imbalance, as it does not distinguish between types of errors.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

Precision indicates how many of the predicted positive instances are actually correct. It is particularly important when the cost of false positives is high.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

Recall measures the ability of the model to identify all relevant instances. It is crucial when missing a positive instance (false negative) is costly.

$$F1 = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}} \tag{4}$$

The F1 score provides a harmonic mean of precision and recall, offering a balanced measure when both false positives and false negatives are important.

$$\text{Macro F1} = \frac{1}{C} \sum_{i=1}^{C} F1_i \tag{5}$$

Macro F1 computes the F1 score independently for each class and then averages them. It treats all classes equally and is useful for evaluating performance on imbalanced datasets.

$$\text{Micro F1} = \frac{2 \cdot \sum TP}{2 \cdot \sum TP + \sum FP + \sum FN} \tag{6}$$

Micro F1 aggregates contributions of all classes before computing the score. It is influenced more by the performance on larger classes and reflects overall system performance.

The evolution of evaluation metrics on the validation set, as presented in Figure 3, provides a more detailed understanding of model performance. Precision, recall, macro F1-score, and micro F1-score all exhibit a rapid increase during the initial training epochs, stabilizing after approximately 100 epochs. The micro F1-score slightly outperforms the macro F1-score, indicating that the model performs consistently well across the dataset, with a slight bias towards more frequently occurring classes.

The stabilization of these metrics suggests that the model reaches its optimal generalization performance relatively early in training. Beyond this point, additional training does not yield meaningful improvements and instead contributes to overfitting, as previously observed in the loss curves.

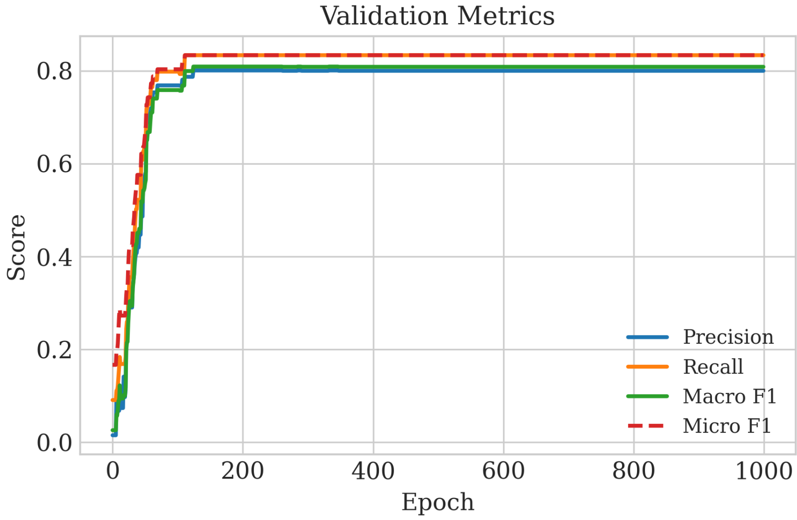

Figure 3: Validation precision, recall, macro F1, and micro F1-score trends over epochs.

3.3 Confusion Matrix Analysis

The normalized confusion matrix shown in Figure 4 provides a class-wise breakdown of prediction performance. A strong diagonal dominance is observed, indicating that the majority of intent classes are correctly classified with high confidence. Several classes, such as *ComparePlacesQuery*, *GetWeatherQuery*, *RequestRideQuery*, *SearchPlaceQuery*, *ShareCurrentLocationQuery*, and *ShareETAQuery*, achieve perfect or near-perfect classification accuracy, reflecting the model's ability to clearly distinguish well-defined intents.

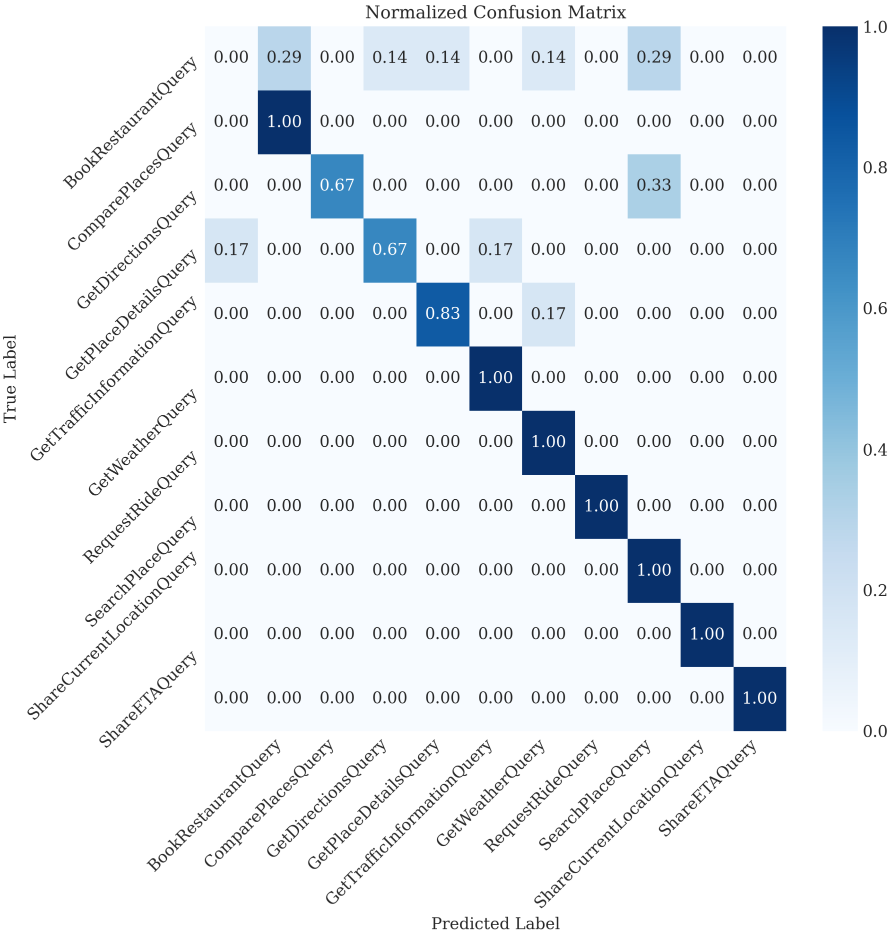

Figure 4: Normalized confusion matrix on the validation set showing class-wise prediction performance.

However, certain classes exhibit noticeable misclassifications. For instance, *BookRestaurantQuery* shows distributed predictions across multiple classes, suggesting ambiguity or insufficient distinguishing features in the input utterances. Similarly, *GetDirectionsQuery* and *GetPlaceDetailsQuery* demonstrate partial confusion with semantically related categories. These misclassifications can be attributed to overlapping linguistic patterns between intents, which pose challenges even for context-aware models like BERT.

3.4 Error Analysis

A closer examination of the results reveals that misclassifications are concentrated among specific intent pairs with overlapping semantic roles. Notably, ambiguity is observed between `BookRestaurant` and `GetWeather`, `PlayMusic` and `SearchCreativeWork`, as well as `GetWeather` and `SearchScreeningEvent`. These pairs share common feature patterns such as location entities, temporal expressions, or content-related keywords, which reduces the discriminative power of the learned representations.

The identification of these ambiguities is performed through analysis of the confusion matrix, where consistently high off-diagonal values between particular class pairs indicate systematic misclassification. Additionally, per-class Precision, Recall, and F1-scores highlight that these classes exhibit comparatively lower scores, confirming that the model struggles to clearly separate them.

In contrast, intents such as `RateBook` and `AddToPlaylist` demonstrate minimal confusion with other classes, as evidenced by strong diagonal dominance in the confusion matrix and consistently high class-wise evaluation metrics. This indicates that classes with more distinct feature distributions are effectively learned by the model.

Overall, the errors are not uniformly distributed but are localized to a subset of semantically related classes, as identified through confusion matrix patterns and class-wise performance metrics.

4 Conclusion

The present study demonstrates the effectiveness of fine-tuning a transformer-based model for intent classification on the SNIPS dataset, achieving strong performance across multiple evaluation metrics. The model exhibits rapid convergence during the initial training phase, as evidenced by the steep improvement in both accuracy and F1-scores within the early epochs. However, the training dynamics also reveal a clear divergence between training and validation performance, with training accuracy approaching near-perfect levels while validation accuracy stabilizes at a lower value. This behavior, along with the increasing validation loss after a certain point, indicates the presence of overfitting, suggesting that the model begins to memorize training data rather than improving generalization.

Despite this limitation, the model demonstrates robust classification capability across most intent categories, as reflected in the strong diagonal dominance of the confusion matrix. Several classes are predicted with near-perfect accuracy, indicating that the model effectively captures well-defined and distinct semantic patterns. The remaining misclassifications are primarily concentrated among semantically similar intent categories, highlighting the inherent ambiguity present in natural language queries rather than deficiencies in model capacity.

The analysis further shows that the model reaches its optimal generalization performance relatively early in the training process, after which additional training yields diminishing returns and contributes to overfitting. This observation suggests that techniques such as early stopping, regularization, or parameter-efficient fine-tuning could be employed to improve generalization. Additionally, incorporating more diverse training samples or contextual information may help reduce ambiguity between closely related intents.

In summary, the fine-tuned BERT model provides a strong baseline for intent classification tasks, achieving high accuracy and balanced performance across classes. While the current approach is effective, future improvements should focus on enhancing generalization and resolving semantic overlaps to further refine classification performance.